```
#importing modules
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

```
#inserting data
data=pd.read_csv('homeprices.csv')
data
```

	area	price
0	2600	550000
1	3000	565000
2	3200	610000
3	3600	680000
4	4000	725000

```
#
x=pd.DataFrame(data['area'])
y=pd.DataFrame(data['price'])
```

```
plt.xlabel('area')
plt.ylabel('price')
plt.scatter(data.area,data.price,color='green',marker='*')
```

```
<matplotlib.collections.PathCollection at 0x28bd3448650>
```


```
#divide data to train and test sets
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2)
```

```
linmodel=LinearRegression()
```

```
linmodel.fit(x_train,y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
linmodel.coef_
```

```
array([[131.73076923]])
```

```
linmodel.intercept_
```

```
array([190961.53846154])
```

```
linmodel.predict(x_test)
```

```
array([[665192.30769231]])
```

```
y_test
```

	price
3	680000

```
linmodel.predict([[1200]])
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py:464: UserWarni
  warnings.warn(
array([[349038.46153846]])
```